TANVIR AHMED

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I am a final-year Computer Science and Engineering student at North South University, working on multimodal machine learning: which inputs need an expensive model, and when a model should say that it does not know. One accepted paper, three under review, and three months as a research assistant at the NSU Trimodal AI Lab.

Education

Bachelor of Science in Computer Science and Engineering **Aug 2022 – Dec 2026**

North South University, Dhaka. Dept. of ECE. CGPA: **2.99 / 4.00**

Higher Secondary Certificate (HSC), Science **Sep 2019 – Dec 2021**

Milestone College, Dhaka. GPA: **5.00 / 5.00**

Experience

Research Assistant, Trimodal AI Lab **Mar 2026 – May 2026**

North South University, Dhaka. Contract, on-site, academic research.

- Worked on multimodal and agentic AI systems under faculty supervision: reading the literature, testing ideas and building prototypes.
- Ran experiments with other lab members and wrote up the results. In the lab since 2025.

Publications

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency **Accepted**

Tanvir Ahmed. *LM-KBC 2026 shared task (5th edition), AKBC at EMNLP 2026.*
Beats the official baseline on five of the six relations, with no retrieval and no fine-tuning.
Code: github.com/minhaz-42/lm-kbc-2026

SimCert: A Classical-Simulability Audit of Trained QNLP Text Models **Under review**

Submitted to QTML 2026.
Whether the quantum component is load-bearing. Code: github.com/minhaz-42/simcert-qnlp

Also under review:

- Observing Is Not Free: The Decision Value of Visual Observations in Vision-Language Agents
- Knowing What You No Longer Know: Calibrated Belief Validity for Persistent World Memory

Research Projects

Frame-Aware Multi-Model Scheduling for Efficient Video Captioning **2026**

Routes video captioning between a 4B and a 27B model so the large one is used only on hard footage. Senior design project, lead author of four. *PyTorch, CLIP, Gemma 3, Qwen3-VL.*

Bangla Mental Health Conversational Agent **Ongoing**

Chatbot for Bangla-speaking users on fine-tuned Gemma-3-Bangla and Qwen-2.5-0.5B-Bangla models.

Technical Projects

- **Pakna Council** (2026). Local-first web app that runs several LLMs in parallel on one prompt through Ollama and puts their answers side by side, with vision workflows and model comparison. *Django, React, Docker, Ollama, WebSocket.* Source · Live site
- **Nestledown** (2026). Cozy 3D exploration game built solo in Godot 4, in which every model, texture, animation and sound is script-generated and no third-party asset enters the repository. A parallel build runs on a custom C++20 engine. *Godot 4, Blender, C++20, SDL3.* Announcement trailer
- **SHAND** (2026). Detects unstated assumptions in English text with an inspectable rule-based system rather than a trained scorer. *Python, NLP, D3.js.* Source · Live site
- **FitWell** (2025). Meal planning, BMI tracking and health insights, with an open-weight model generating plans from a user's goals. *Django, Qwen, SQLite.* Source

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C/C++, PHP, SQL, HTML, CSS, LaTeX
Machine learning: Computer vision, NLP, LLMs, vision-language models, calibration, quantum ML
Frameworks: PyTorch, PyTorch Geometric, scikit-learn, Transformers, OpenCV, CLIP, PennyLane
Software and systems: Django, React, Node.js, REST APIs, Docker, CI/CD, SQL, Git, Linux, STM32

Extracurricular Activities

IEEE NSU Student Branch, General Member **2023 – Present**

Technical workshops and seminars run by the branch, and its community outreach programmes.

NSU Trimodal AI Lab, Reading Group **2025 – Present**

Reads and discusses multimodal and agentic AI papers with other members of the lab.

References

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